

Mateusz Kapusta

Software Engineer · ML Engineer

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Experience

Machine Learning Intern

Veldhoven, NL

ASML NV

October 2025 - Jun 2026

- Developing Bayesian ML tools coupled to the physical simulations for better diagnostic of NXE machines. Applied research in the field of simulation-based inference. Working on various tasks in different domains including timeseries modelling, bayesian estimation, volumetric rendering.
- Created methods was selected as the most promising candidate and is earmarked as the potential company-wide tool for diagnostic of contamination.
- High performance implementation of several algorithms, multi GPU training and inference, code optimization, performance tuning.

Quantitative Developer Intern

Wrocław, PL

UBS AG

Jul 2025 - Sep 2025

- Benchmarking, optimizing and productionizing the C/C++ multithreaded Monte Carlo simulation code that is a core component of the Lombard Loan pricing engine.
- Parallel optimization of the code with the help of the IntelTBB library.
- Final code optimization resulted in 3x speedup and >2x memory saving.
- Resolving deployment issues in the cloud during the roll-out.
- Developing front-end of the engine written in R, preparing SQL queries for data loading.

Skills

Programming

C++, Python, Go, R, Julia

Languages

Spoken Languages

English (C1): 103 TOEFL, Polish (Native), German (B1/B2)

Skills

MPI, CUDA, Torch, Jax

Education

Warsaw University

Warsaw, PL

M.S. in Astrophysics, minor in CS

August 2023 - May 2026

- Physics specialization: Computational Astrophysics, Machine Learning in Physics
- CS specialization: Probabilistic ML, High-Performance Computing.

Warsaw University

Warsaw, PL

B.S. in Astrophysics, minor in CS and Math

August 2020 - May 2023

- Physics specialization: Observational Astrophysics, Machine Learning in Physics.
- CS specialization: Machine Learning, Probabilistic ML.
- Math specialization: Stochastic processes.

Awards

- Prize for scientific research granted by the Minister of Education 2025
- University Physics Competition - part of gold-medal winning team representing Faculty of Physics
- Various results in high school science olympiads: polish representation for IOAA 2019 & 2020, finalist in math, physics and astronomy olympiads.